温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Introduction to Business Analytics

Dawei Chen

Week 1-2, 04 Sep 2025

Outline

- What is Business Analytics? – Definition
- Why Business Analytics? – Decision-Making
- Types of Business Analytics – Spectrum & Examples
- Challenges in Business Analytics
- Summary

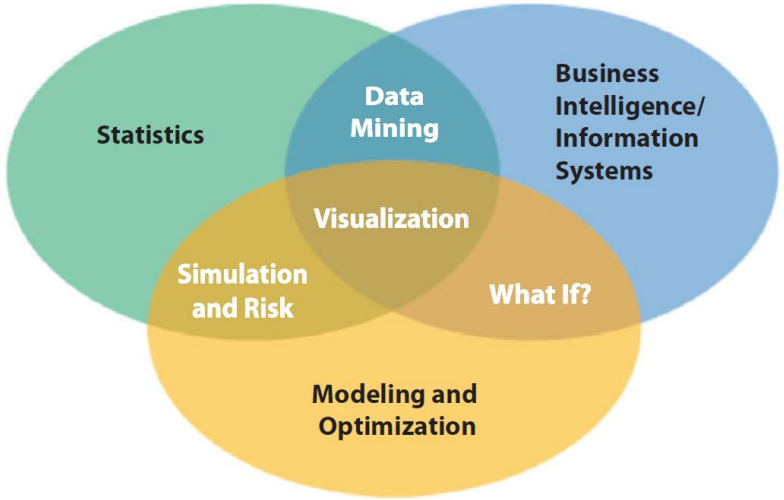
Chapter 1, Business Analytics, 2nd Edition, James Evans, Person 2017.

Definition of BA

Business Analytics is the use of:

- Data
- Information technology
- Statistical analysis
- Quantitative methods
- Mathematical or computer-based models

to help manager gain improved insight about their business operations and make better, fact-based decisions.



A visual perspective of BA

Chapter 1, Business Analytics, 2nd Edition, James Evans, Person 2017.

Why BA? Who Uses It?

Levels of Decisions in Firms:

- *Strategic:* high-level, set overall goals (3-5Y)
- *Tactical:* mid-level, define strategies to achieve goals (1Y)
- *Operational:* day-to-day, ensure smooth business operations

Common Approaches to Decisions

- Tradition
- Intuition
- Rules of thumb
- **Data-driven analysis** ✓

Decision-Making Process:

- Identify and define the problem
- Set evaluation criteria
- Generate alternative solutions
- Evaluate options
- Select the best solution

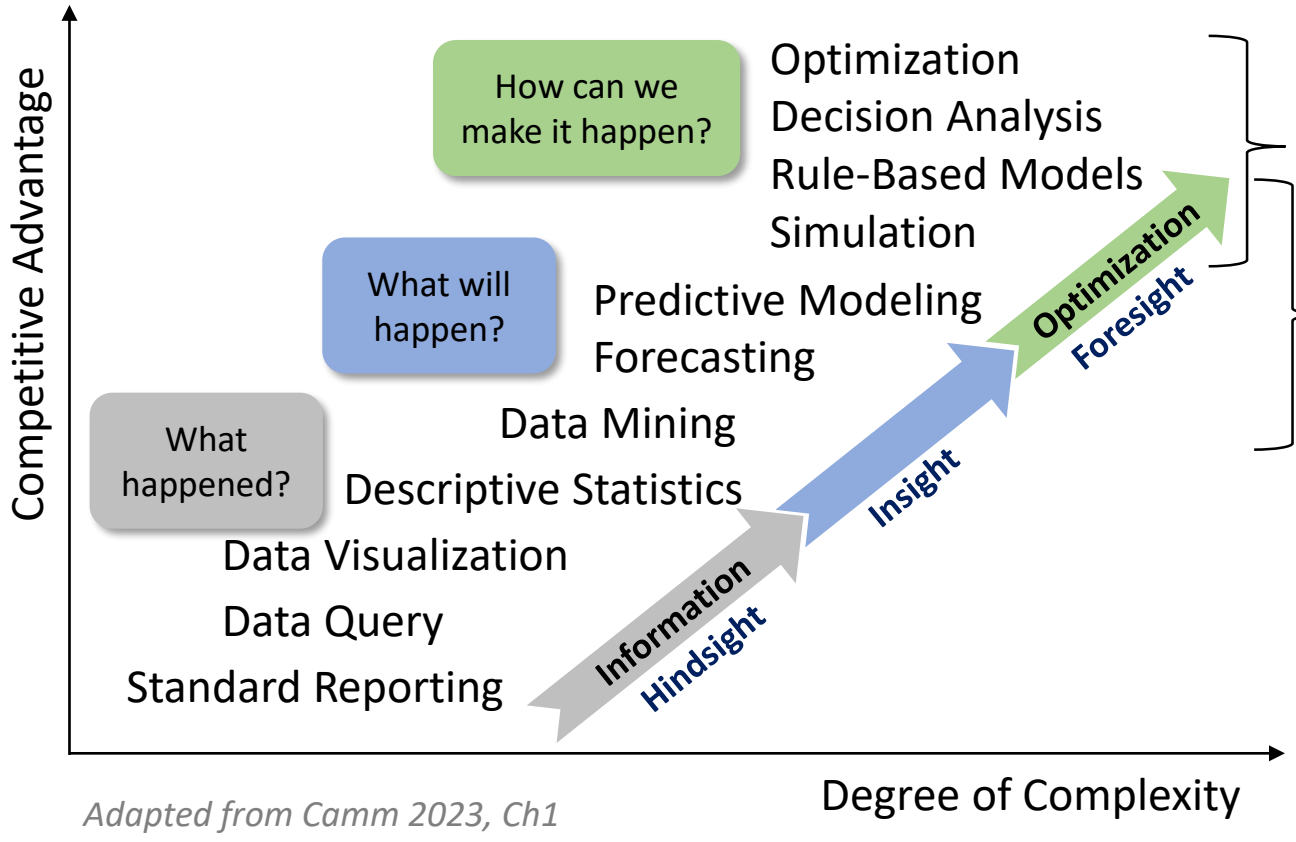
Uncertainty

⇧

Data Insights

Spectrum of BA

Diagnostic Analytics: why did it happen? Builds on descriptive results by finding correlations, causes, and contributing factors.



Adapted from Camm 2023, Ch1

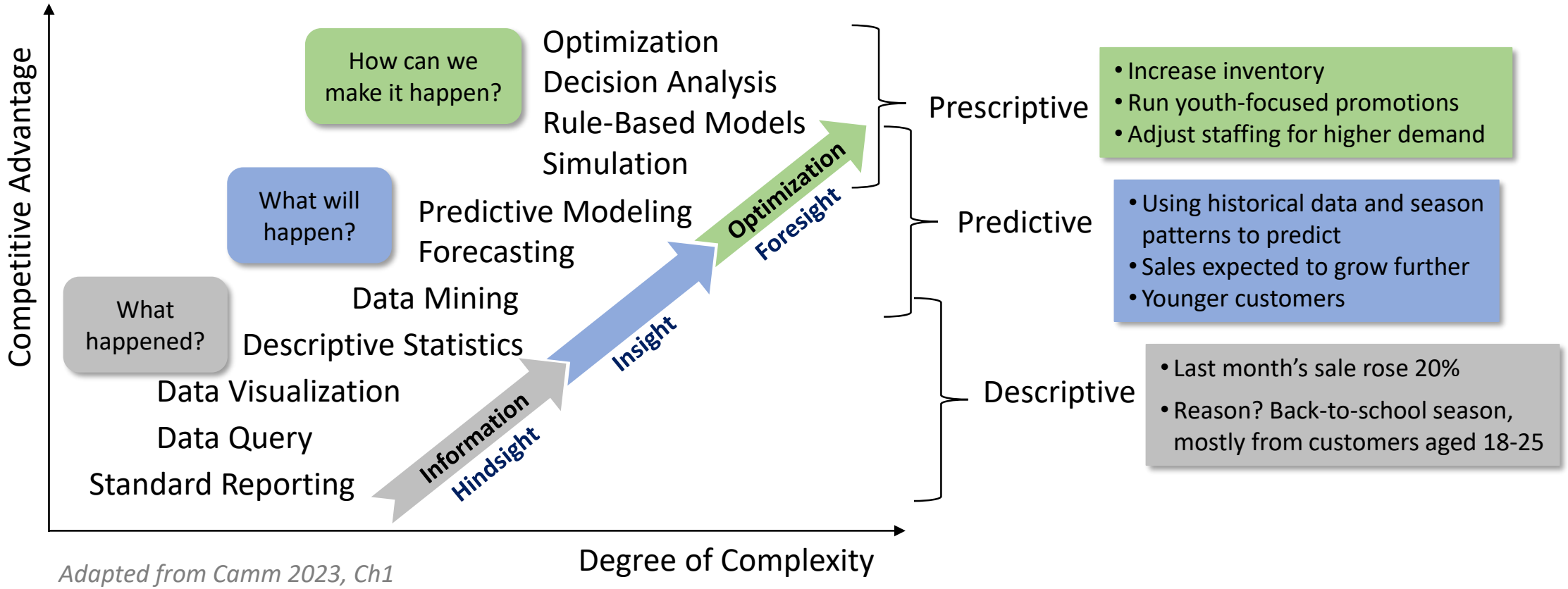
Prescriptive — Identify the best alternatives to minimize or maximize some objectives

Predictive — Predict future by examining historical data, detecting patterns or relationships in these data, then extrapolating these relationships forward in time

Descriptive — Use of data to understand past & current business performance & make informed decisions

Exploratory Data Analysis (EDA): a starting point, uses visualization and summary statistics to explore patterns, spot anomalies, and generate hypotheses.

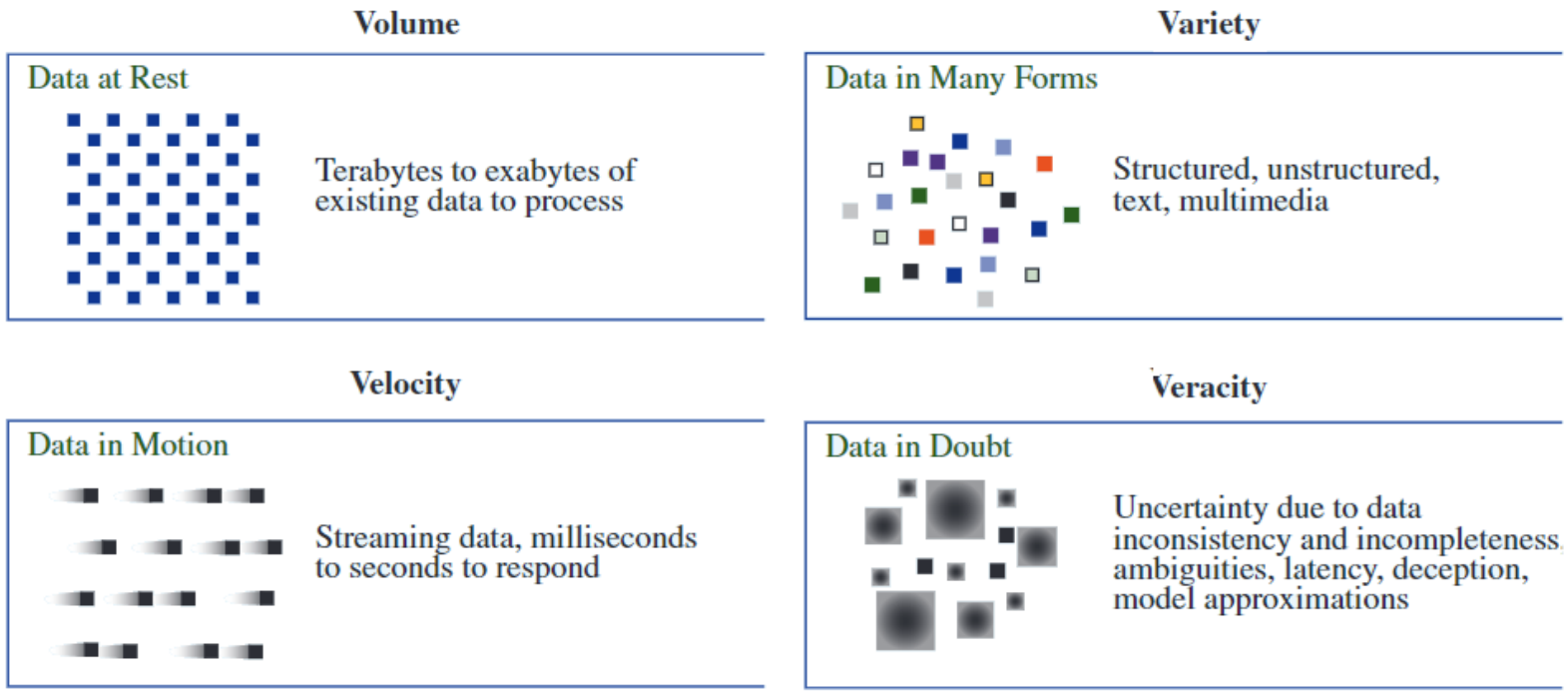
Example: A Sneaker Factory in Wenzhou



Adapted from Camm 2023, Ch1

Challenges in BA – Big Data

- Massive amounts of business data from a wide variety of sources, much of which is available in real time, and much of which is uncertain or unpredictable.
- Size is a “moving” target, depending on available technologies capabilities.
- IBM 4 Vs:



Source: IBM, adapted by Camm 2023, Ch1, p.7

Metrics and Data Classification

- Metric: a unit of measurement that provides a way to objectively quantify performance
- Measurement: act of obtaining data associated with a metric
- Measures: numerical values associated with a metric

Eg: Let's assess the performance of a retail store.

Metric
• Net profit • ROI • customer satisfaction • market share growth

Measurement
• Identifying data sources • Collecting data

Measure
• Pre-validated measures • Create and validate new measures

Types of Metric

- **Discrete metric**
  - one that is derived from counting something.
  - Examples: a delivery is either on time or not; an order is complete or incomplete; or an invoice can have one, two, three, or any number of errors.
  - Some discrete metrics would be the proportion of on-time deliveries; the number of incomplete orders each day, and the number of errors per invoice.
- **Continuous metric**
  - based on a continuous scale of measurement.
  - Examples: Any metrics involving dollars, length, time, volume, or weight, for example, are continuous.

Measurement of Scales

- Categorical (nominal): classified into distinct categories (no order)
- Ordinal: ordered or ranked categories (relative order, no exact distance)
- Interval: ordinal, constant differences between observations, arbitrary zero points
- Ratio: continuous, equal intervals, a natural zero

	A	B	C	D	E	F	G
1	**Purchase Orders**						
2							
3	**Supplier**	**Order No.**	**Item No.**	**Item Description**	**Item Cost**	**Quantity**	**Order Date**
4	京东方	Sep11001	1122	面板	$99	100	04/09/2025
5	赛腾股份	Sep11002	1243	苹果智能(AI)	$88	101	04/09/2025
6	蓝思科技	Sep11003	5462	玻璃盖板	$77	102	04/09/2025
7	歌尔股份	Sep11004	9527	声学器件/马达	$66	103	04/09/2025
	Categorical	Ordinal	Categorical	Categorical	Ratio	Ratio	Interval

Calendar date?
Time of day? 6:00pm
Duration? 5mins

Challenges in BA – Cloud Computing & Security

- The massive amounts of available data have led to numerous innovations that help make the data useful for decision-making.
  - Cloud computing ("the cloud") refers to the use of data and software on servers housed externally to an organization via the internet.
  - The cloud has made storing and processing massive amounts of data feasible and cost-effective.
- The security of highly confidential data stored on the cloud is critical to companies and must be protected from hackers.
  - Data security protects stored data from destructive forces or unauthorized users.

Challenges in BA – Privacy

- Increased attention has been paid to ethical concerns around data privacy and the ethical use of models based on data.
  - Clients and customers must understand the trade-offs between allowing their data to be collected and the benefits they accrue from allowing a company to collect and use that data.
  - An agreement must be signed between the customer and the company.
- Companies have an obligation to uphold that agreement and make every effort to ensure that the data are protected from data breaches.
  - Data breaches, defined as unauthorized uses of data, are a major concern for all companies.

Challenges in BA – Privacy Compliance

- The General Data Protection Regulation (GDPR), one of the strictest privacy laws that went into effect in the European Union in May 2018, stipulates that
    - the request for consent to use an individual's data must be easily understood and accessible,
    - the intended use of data must be specified and easy to withdraw consent, and
    - the individual has a right to a copy of their data and the right to demand their data be erased.
- It is the responsibility of analytics professionals to understand the laws associated with collecting, storing, and using individuals' data.

AI in BA: The Good

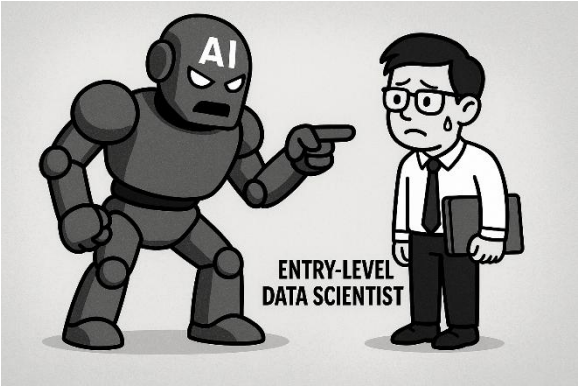
- **Efficiency:** Automates repetitive tasks (data cleaning, model tuning)
- **Scalability:** Handles large and complex datasets quickly
- **Accessibility:** Lowers technical barriers with AI coding assistants (e.g., ChatGPT, Copilot)
- **Insight Generation:** Enhances predictive analytics and prescriptive recommendations
- **Visualization:** Creates smarter, automated dashboards and reports
- **Decision Support:** Provides faster, more informed business decisions

Generated by ChatGPT 5

AI in BA: The Bad

- **Bias & Fairness:** Risk of reinforcing biased data or flawed models
- **Over-Reliance:** Managers may trust AI outputs without critical evaluation
- **Transparency:** “Black box” models reduce interpretability
- **Data Privacy:** Use of sensitive data raises ethical and legal concerns
- **Skill Gaps:** Analysts risk losing core statistical and analytical skills
- **Costs:** Investment in AI tools, infrastructure, and talent can be high

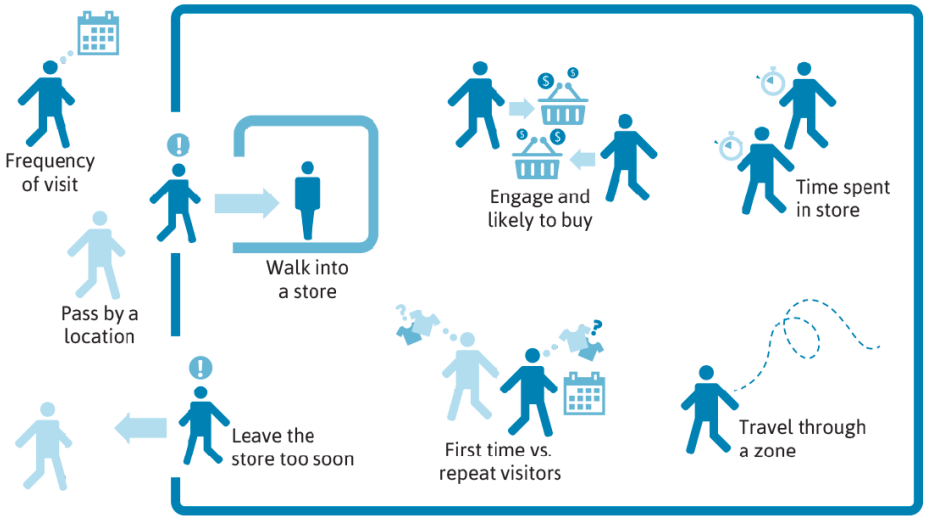
- *What is the impact of AI on business analytics labor market? Demand, replacement, upskilling, deskilling?*
- *How to start and prepare yourself for BA career in era of AI?*



Generated by ChatGPT 5

Summary

- Business analytics supports better-informed decisions
- Three categories of analytics
  - descriptive, predictive, and prescriptive
- Metric and data classification
  - Metric, measurement, measure
  - Categorical, ordinal, interval, ratio
- Challenges in business analytics
  - Big data, security, privacy, AI



Eg: Customer Behavior in Retail Store
- What data can we collect?
- What analysis can we perform with it?

Closing

- Topics for Next Week
  - Jupyter Notebook basics, Python basics review
  - NumPy & Pandas essentials
- If You're New to Python
  - Spend 1-2 hours to get familiar with syntax
  - Quick tutorials: W3Schools / 菜鸟教程
  - Key point: **Don't be afraid of Python!**
- Learning Path *(my experience)*
  - Start with Python basics (菜鸟教程)
  - Use *Python for Data Analysis* by Wes Mckinney
  - Learn by doing: practice, error, debug, and improve
  - (Optional, advanced) Independently complete a data analysis project

No need to worry—this is a hands-on, step-by-step data analysis course. Together we will explore the beauty, creativity, and a few practical "hacks" (dirty tips, 邪修) of data analysis.

Kaggle example:
EDA to Prediction(DieTanic)